Assignment 3

From Clinical Notes to Optimal Service Policies

Westfield Children's Centre — Applied Case Study

Case Overview: Westfield Children's Centre

Sarah Chen, Program Manager of the Early Years Program, is heading into September 2024 with a growing waitlist, fixed clinical capacity, and a strict eligibility boundary: children leave the program at school entry.

To improve planning, Chen introduced a standardized pathway of two six-session blocks with an expected course of twelve sessions. The change reduced very long courses of care, but it may also have pulled some cases closer to the full pathway even when earlier discharge was possible. One senior speech-language pathologist, David Patel, stood out for regularly discharging children earlier than most of his peers. The scheduling data showed when his cases ended, but not why.

Patel believed the answer was in the clinical notes. Read over time, those notes describe how each child's care is unfolding—the goals being targeted, the level of independence achieved, and how both change across sessions. Patel has labeled a subset of notes for patterns of progress over time, and those examples will serve as the basis for evaluating other clients' clinical records.

Your task is to design the analytics pipeline, starting from raw notes, that could turn this information into signals for planning service needs and capacity use.

Data Files Provided

- `labeled_notes.json` hand-scored session notes from 40 clients, used for Q1 validation
- `unlabeled_notes.json` 80 clients × 12 session notes to be scored by your prompt
- `client_features.csv` intake characteristics for all historical clients
- `waitlist.csv` intake characteristics for clients currently waiting to begin service

Q1 Prompt Engineering: Extracting Per-Session Progress Scores

Context

When David Patel reviews a child's record, he does not judge progress from a single note in isolation. Instead, he reads the notes in sequence and evaluates how much progress occurred from one session to the next. This judgment is grounded in the goal-level framework from **Exhibit A**. Movement to a higher goal level, reduced reliance on prompting, and greater consistency or spontaneity are all indicators of progress over time.

Your task is to encode that reasoning into a prompt. The model will receive a client's ordered session notes and must compare each note with the one immediately before it. For every consecutive pair of notes, the model should assign a progress score from 0 to 3 reflecting the child's relative progress between those two sessions.

Rating Scale

Score	Label	When to use
0	**Maintenance / minimal change**	The child is functioning at essentially the same level as before. Accuracy, cueing needs, and hierarchy level remain similar.
1	**Small but clear improvement**	Modest progress within the same general level—slightly better consistency, less cueing, or improved carryover—without a major increase in independence or hierarchy level.
2	**Meaningful clinical progress**	An obvious step forward that matters clinically: moving from inconsistent to fairly consistent performance, clearly requiring less support, or showing broader generalization across activities.
3	**Major gain / step up in level**	A clear breakthrough: a hierarchy jump, major gain in independence, or a new level of spontaneous use. **Use sparingly.**

(a) Prompt Development and Validation

Develop a prompt that takes a client's ordered sequence of session notes as input and returns a progress score for each consecutive pair. For example, if a client has notes for Sessions 1 through 12, the model should return a score for each of the following transitions:

- Session 1 → Session 2
- Session 2 → Session 3
- ...
- Session 11 → Session 12

Begin with an initial version of your prompt and run it on `labeled_notes.json`. You may experiment with strategies such as:

- Persona / role assignment (e.g., "You are a senior SLP evaluating a child's therapy record...")

- Explicit score definitions referencing the rating scale above
- Few-shot examples drawn from the labeled set
- Output format constraints (e.g., return only a JSON object `{"scores": [2, 1, 0, ...]}`)
- Negative examples or boundary clarifications (e.g., distinguishing a score of 1 from a score of 2)

Because these scores are judgment-based and ordinal, choose and justify an evaluation approach that you believe best reflects prompt quality for this task.

Please report:

- The LLM used
- The final refined prompt, including the strategies you implemented
- The evaluation approach, with a brief justification
- The resulting performance
- The confusion matrix, with a brief interpretation of what it reveals about the model's behavior

(b) Large-Scale Scoring

Apply your validated prompt to every client in `unlabeled_notes.json`. Save the output as `scored_notes.csv` with columns: `client_id`, `session`, `score`.

Q2 Trajectories, Optimization & Selecting K

Context

You now have a per-session progress score for every client in the dataset. Taken together, these scores form a progress trajectory for each child across care. Some children show strong gains early and then level off. Others improve gradually. Others show uneven or delayed progress.

This question asks you to do three things in sequence: build a clustering model, derive an optimal reassessment policy from the clusters it produces, and then use what you learn from that policy to decide how many clusters to use. reveals.

(a) Build the Clustering Model

Build a pipeline that converts per-session scores to a 0–3 scale, constructs cumulative progress trajectories across Sessions 1–12, and applies K-means clustering for a value of K that you supply. Generate spaghetti plots showing individual trajectories within each cluster and the cluster mean.

(b) Derive the Optimal Reassessment Policy

For each cluster, apply the newsvendor-style audit model below to the clusters you have built. This is the **prescriptive** core of the question: given a cluster's distribution of stopping points, when should Westfield schedule the reassessment?

Model setup

Parameter	Definition
$T_{\max}$	Maximum sessions in the funded pathway = 12
t_i^*	Stopping point for client i: earliest session where cumulative progress $\geq 90\%$ of that client's total cumulative progress across all sessions
Q	Reassessment session chosen for a given cluster (the decision variable)
$F_c(Q)$	Empirical CDF of t^* within cluster c: $F_c(Q) = P(t^* \leq Q \mid c)$

One-shot audit rule (mirroring Westfield's re-assessment logic in Exhibit B):

- If $Q < t_i^*$: reassessment is too early; the child continues to $T_{\max}$ (no sessions saved)
- If $Q \geq t_i^*$: plateau confirmed; care ends at Q ($T_{\max} - Q$ sessions saved)

Expected sessions saved: $E[\text{savings}](Q) = F_c(Q) \times (T_{\max} - Q)$

Objective: for each cluster c, find $Q_c^* = \arg\max_Q \; E[\text{savings}](Q)$

For each client, compute t_i^* using the fraction-of-total-progress rule. Then, for each cluster and for $Q = 1, 2, \ldots, 12$, compute $E[\text{savings}](Q)$ and identify Q_c^*. Do this for each value of K you are evaluating.

(c) Select K

Use the policy outcomes to commit to a final value of K. The goal is a clustering that produces reassessment policies that are genuinely distinct and actionable.

State your final choice of K and justify it in terms of both what the trajectory shapes look like and what the optimization revealed.

(d) Required Plots for Your Final K

Using your chosen K, produce the following three plots:

Plot 1 — t^* distributions. For each cluster, plot a histogram of t_i^* values (one panel per cluster, shared x-axis). The x-axis is the stopping session (1–12) and the y-axis is the count of clients. These distributions show whether stopping points are concentrated or spread out within each group, and whether they tend to fall early or late in the pathway.

Plot 2 — Expected sessions saved vs. Q. For each cluster, plot $E[\text{savings}](Q)$ against $Q = 1, 2, \ldots, 12$ on a single set of axes (one line per cluster). Mark Q_c^* for each cluster with a vertical dashed line. This curve makes the tradeoff between auditing early and auditing late visible, and shows why different clusters call for different reassessment timing.

Plot 3 — Optimized Q^* vs. mean(t^*) baseline. For each cluster, compute a naive baseline policy by rounding the mean t^* within that cluster. Plot expected sessions saved per child side-by-side for the optimized Q^* and the baseline (one grouped bar per cluster). Report the overall difference in expected savings across all clients and explain intuitively why the optimized policy can outperform the mean heuristic.

(e) Summary Table and Cluster Labels

Fill in the table below for your final K:

Cluster	Size of Cluster	Q_c^*	E[saved/child]	% of sessions saved
Total	—		—	

(f) Implications for the Case

Chen and Patel suggest that clients may receive similar numbers of sessions but still progress very differently through care. To what extent do your clusters support this claim? What do the trajectory patterns and the differentiated reassessment policy suggest for how Westfield plans and schedules care?

Q3 Predictive Analytics: From Intake Profiles to Expected Service Demand

Context

The clients analyzed in Q2 are historical clients. Their progress trajectories are now known because they have already moved through the pathway. Chen's planning problem, however, is forward-looking. She must make decisions about children who are still on the waitlist, before any session notes exist.

Your task is to use the historical clients from Q2 to learn how intake characteristics relate to different trajectory types. If age, referral reason, complexity, or other intake features are associated with different progress patterns, Westfield may be able to predict which type of pathway a new referral is most likely to follow.

The goal of this question is to predict the likely trajectory type of each new referral using information available at intake. Those predictions feed back into Q2(g) to estimate the capacity impact of the differentiated reassessment policy on the current waitlist.

(a) Exploring Intake Features by Trajectory Type

Using `client_features.csv`, compare intake characteristics across the trajectory clusters you identified in Q2. Produce visual summaries of how these features differ across clusters. Then answer:

- Which features appear most useful for distinguishing trajectory groups?
- Which features overlap substantially across groups?

(b) Train a Model to Predict Trajectory Group

Using the historical clients from Q2, train two classifiers to predict trajectory cluster membership from intake features alone:

1. A parametric model, such as multinomial logistic regression
2. A more flexible non-parametric model, such as a random forest or gradient-boosted trees

Use a stratified 80/20 train/test split. For each model, report accuracy and the confusion matrix

State which model you would recommend for this setting and why.

(c) Predict Trajectory Mix for the Waitlist

Apply your best model to the clients in `waitlist.csv` to estimate which trajectory group each waiting client is most likely to follow.

Using those predicted assignments together with the cluster-level Q_c^* values from Q2, estimate the total session capacity Westfield would need to serve the current waitlist under the differentiated reassessment policy. Compare this against the baseline assumption that every client receives all

$T_{\max} = 12$ sessions. Specifically:

- For each waiting client, assign the expected sessions delivered under the policy for their predicted cluster (i.e., $E[\text{delivered} \mid Q_c^*]$)
- Sum across all waiting clients to obtain a total expected session demand
- Report how this compares to the $T_{\max}$ baseline (total sessions if no differentiated policy were applied)
- Identify which trajectory groups drive the most capacity savings and which contribute little